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Isochrones in primordial magnetic field evolution

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ABSTRACT

Context. In the early Universe, a primordial magnetic field undergoes a turbulent decay while its length scale increases due to an inverse cascade. The size of the largest processed eddy scales with the Alfvén speed and grows with time. In a diagram of Alfvén speed versus length scale, all possible solutions must lie on a line through the origin, with a slope proportional to the inverse of the present time. In principle, however, such lines can also be defined for earlier times.
Aims. The lines for earlier times form isochrones that might be observationally accessible, for example, through the magnetically driven stochastic gravitational wave background. However, the position and slope of these isochrones is sensitive to the zeropoint of the time. We show that for any initial magnetic field, a proper time can be determined such that the resulting isochrones at early times are nearly parallel to those at late times, that is, they have the same slope.
Methods. We used two-dimensional numerical simulations of decaying magnetohydrodynamic turbulence and varied the initial position of the peak in the magnetic energy spectrum. In this case, the evolution is governed by the conservation of anastrophy.
Results. A fit to the Alfvén time yielded an accurate estimate of the factor by which the decay time is longer than the Alfvén time, while the offset in the fit provided an estimate of the proper time that needs to be added to the nominal time since the beginning of each simulation. We also found that an initial velocity field with a realistic strength produced a straighter track from the beginning.
Conclusions. The magnetic field parameters lie on universal isochrones even for early times. They provide a testable framework for magnetic fields generated at times as early as the end of inflation, starting with the time of reheating.

Key words. magnetohydrodynamics (MHD) – turbulence – cosmology: miscellaneous

1. Introduction

Primordial magnetic fields can still be observable in the voids of the cosmic web, while the pollution of magnetic fields by purely astrophysical (i.e., galaxy evolution-driven using expulsion of magnetized winds and/or jets) mechanisms are thought to be incapable of producing large enough filling factors of the magnetic field, as seen in numerical simulations (e.g., Donnert et al. 2010; Bondarenko et al. 2022; Tjemsland et al. 2024; Vazza et al. 2025; Ghosh et al. 2026). At the simplest level, primordial magnetic fields are characterized by their typical strengths and length scales. A useful measure of their rms value $B_{\rm rms}$ is the Alfvén speed $v_A = B_{\rm rms}/\sqrt{\mu_0\rho}$, with ρ being the density and μ_0 the vacuum permeability: it quantifies the speed at which a magnetic field perturbation spreads. Thus, after a certain time t, the typical radius of such a perturbation is $\sim v_{\rm A}t$.

Primordial magnetic fields are generated on scales smaller than the horizon scale at the epoch of generation (e.g., Grasso & Rubinstein 2001). When the generation occurs during or before inflation, however, the maximum scales can be larger than the present-day horizon (e.g., Grasso & Rubinstein 2001; de Souza & Opher 2008; Widrow et al. 2012; Durrer & Neronov 2013; Subramanian 2016). The Universe expands, but this expansion is commonly scaled out, and so we talk about comoving length scales and comoving magnetic field strengths to account for the dilution of the field with time. For example, a physical magnetic field of 10^{24} G at the electroweak epoch corresponds to a comoving field of 1 μG today, and the horizon scale of about 1 cm corresponds to 6×10^{-4} pc (Kahniashvili et al. 2013). The actual magnetic field is weaker still because of turbulence, however, which speeds up the decay of the field. However, as the magnetic field undergoes a turbulently enhanced decay, it can still grow on large length scales (Brandenburg et al. 1996). Thus, the magnetic energy spectrum increases at small wave numbers through what is called an inverse cascade (Frisch et al. 1975; Pouquet et al. 1976).

The speed at which the turbulent eddies grow is expected to be limited by $v_{\rm A}$. Kahniashvili et al. (2013) talked about the largest processed eddy whose size is $\sim v_{\rm A}t$. This leads us to the expectation that in our Universe, whose age is $t_{\rm U}\approx14$ Gyr, there should be no causally generated turbulent magnetic fields whose characteristic length scale exceeds $v_{\rm A}t_{\rm U}$.

Using $\rho=4\times10^{-31}$ g cm^{-3} for the baryon density and $B_{\rm rms}=1$ nG for the magnetic field, we find $v_{\rm A}=4$ km s^{-1}, and therefore the largest processed eddy has a size of 60 kpc. For a weaker field, $v_{\rm A}t$ is correspondingly smaller. In a diagnostic diagram of magnetic field strengths or $v_{\rm A}$ versus characteristic length scale

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ξ_M, the above considerations lead to a line $v_A = \xi_M(t_U)/t_U$ below which no causally generated turbulent magnetic fields should ever be detected within the time t_U (Banerjee & Jedamzik 2004). If any magnetic fields below this line are detected, this would mean that they can only be of nonturbulent nature, that is, their large length scales were already acquired at the time of generation and might have resulted from the epoch of inflation.

The existence of a line $v_A = \xi_M(t_U)/t_U$ is quite remarkable because it means that all observed combinations of v_A and ξ_M must lie on this line, which is independent of the initial conditions. Even for very weak magnetic fields, the current value of v_A would still lie on this line, but the point would just be shifted further to the lower left along this line. This remarkable realization of (Banerjee & Jedamzik 2004) was emphasized by Hosking & Schekochihin (2023), who wrote that its great utility has perhaps not been spelled out explicitly, and that one need not know its functional form during the early stages of the decay in order to compute B and ξ_M at later times.

Using numerical simulations and theoretical considerations, we can address the question of where in the diagnostic diagram the parameters v_A and ξ_M of the Universe lie at any earlier time. We thus expect the v_A versus ξ_M diagnostic diagram to be occupied by isochrones given through

$$v_A(\xi_M) = \xi_M(t_*)/t_*, \tag{1}$$

where t_* denotes the time of the isochrone. These isochrones would then all be parallel to each other.

Over the past few decades, many numerical simulations have been performed, where the initial field obeyed a certain magnetic energy spectrum $E_M(k)$, with a peak at the wave number $k = k_B$ and a field strength B_0 related to the integral of $E_M(k)$. This implies that in the v_A versus ξ_M diagram, any point would represent a valid initial condition. This seems to contradict the idea of universal isochrones that are given by $\xi_M(t_*)/t_*$ because it would restrict the permissible initial conditions to lie on the line $\xi_M(t_*)/t_*$ in the v_A versus ξ_M diagram.

Hosking & Schekochihin (2023) have emphasized that the early stages do not matter for computing v_A and ξ_M at later times. This could address the puzzle about the permissible initial conditions. However, it would leave open the question of when the theoretical isochrones become parallel to those at late times and how they are reached in practice.

One question would be whether the ideas of the theoretical isochrones can be applied already to the moments shortly after the end of inflation, for example. This question is relevant when we compare different magnetogenesis scenarios, where magnetic fields might be produced over a range of magnetic energy and length scales. The definition of isochrones is then particularly sensitive to the zeropoint of the time axis, which is a priori a matter of convention. In view of the fact that early Universe magnetic fields can imprint signatures on the stochastic gravitational wave background (Caprini & Durrer 2001, 2006; Kahniashvili et al. 2005, 2008), the definition of isochrones from such epochs can also be observationally relevant.

We demonstrate that to compare different magnetogenesis scenarios, we must first reset the nominal time by a certain amount such that the initial value of $v_A t$ agrees with the initial scale ξ_M. This can be done by determining a fit of the instantaneous Alfvén time versus t, as we discuss in this paper.

In addition to the uncertainty in the zeropoint of the time axis, there is a well-known problem with the use of the Alfvén time itself. It has been found numerically that the actual decay time of magnetohydrodynamic (MHD) turbulence exceeds the Alfvén time by a certain Lundquist-number-dependent factor (Bhat et al. 2021), which we call C_M. In this case, the actual isochrones are expected to lie on somewhat higher lines than the Alfvénic isochrones (see Hosking & Schekochihin 2023). The factor C_M was found to depend on the Lundquist number Lu, but to be saturated at a value of around 50 (Brandenburg et al. 2024). On theoretical grounds, we might expect there to be an additional dependence on the magnetic Prandtl number (Hosking & Schekochihin 2023). This, however, hinges on the validity of earlier findings that the reconnection rate depends on the magnetic Prandtl number (Comisso et al. 2015), which has recently been challenged (Kumar & Brandenburg 2026). We were able to avoid the uncertainty related to the value of C_M as well. This factor automatically emerges from the aforementioned fit of the instantaneous Alfvén time to the nominal time t.

Following earlier work (Brandenburg et al. 2024), we focused on the 2D case. It is different from the 3D counterpart in that the governing conserved quantity is here the anastrophy and not the mean magnetic helicity density in the helical case or the Hosking integral in the non-helical case. However, our goal was to clarify the proper definition of the isochrones, and this can equally well be studied in the simpler 2D case. Furthermore, in 2D, we can reach a wider dynamical range and therefore reduce some artifacts resulting from the finite cutoffs at small and large wave numbers of the domain.

2. The model

2.1. Basic equations

We performed 2D simulations in Cartesian coordinates, (x, y, z), in a 2D domain of size L^2. The magnetic field was assumed to lie entirely in the xy plane and can therefore be written as $\boldsymbol{B} = \boldsymbol{\nabla} \times (\hat{\boldsymbol{z}} A_z)$. The induction equation then reduces to

$$\frac{\mathrm{D}A_z}{\mathrm{D}t} = \eta \nabla^2 A_z, \tag{2}$$

where $\mathrm{D}/\mathrm{D}t = \partial/\partial t + \boldsymbol{u} \cdot \boldsymbol{\nabla}$ is the advection operator, $\boldsymbol{u}$ is the velocity, and η is the magnetic diffusivity. In the limit $\eta \to 0$, Eq. (2) obeys the conservation of anastrophy, $I_A \sim \langle \rho A_z^2 \rangle = \mathrm{const}$ (Fyfe & Montgomery 1976; Pouquet 1978, 1993), where angle brackets denote volume averaging.

We also solved the equations for $\boldsymbol{u}$ and the logarithmic density $\ln \rho$,

$$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} = \frac{1}{\rho} \left[\boldsymbol{J} \times \boldsymbol{B} + \boldsymbol{\nabla} \cdot (2\rho\nu \mathsf{S}) \right] - c_s{}^2 \boldsymbol{\nabla} \ln \rho, \tag{3}$$

$$\frac{\mathrm{D}\ln\rho}{\mathrm{D}t} = -\boldsymbol{\nabla} \cdot u. \tag{4}$$

In Eq. (3), $\mathsf{S}_{ij} = (\partial_i u_j + \partial_j u_i)/2 - \delta_{ij} \boldsymbol{\nabla} \cdot u/3$ are the components of the rate-of-strain tensor S, ν is the viscosity, c_s is the isothermal sound speed, and $\boldsymbol{J} = \boldsymbol{\nabla} \times \boldsymbol{B}/\mu_0$ is the current density. We applied periodic boundary conditions in all directions.

2.2. Initial conditions

Our initial magnetic and velocity fields are given in Fourier space (indicated by a tilde) by $\tilde{A}_z(\boldsymbol{k}) = B_0\, g(\boldsymbol{k})\, S_B(k)$ and $\tilde{\psi}_z(\boldsymbol{k}) = u_0\, g(\boldsymbol{k})\, S_u(k)$, respectively, where $g(\boldsymbol{k})$ is the Fourier transform of a δ-correlated vector field with Gaussian fluctuations, $\tilde{\psi}_z$ is the Fourier transform of the stream function of the velocity $u = \boldsymbol{\nabla} \times (\psi_z \hat{\boldsymbol{z}})$, and k_v is the initial wave number of the energy-carrying eddies for the field v, where $v = B$ or u for the magnetic

and velocity fields, respectively. Likewise, $S_v(k)$ with $v = B$ or u determines the spectral shape with (Brandenburg 2020)

$$S_v(k) = \frac{k_v^{-d/2}(k/k_v)^{(\alpha_v^{\rm sub}+d-1)/2}}{[1+(k/k_v)^{2(\alpha_v^{\rm sub}-\alpha_v^{\rm ine})}]^{1/4}}\, e^{-(k/K_v)^{n_v}}, \quad (5)$$

where $d = 2$ is the dimension for our 2D runs, and K_v determines the wave number of an exponential cutoff. In all cases, we used $K_B = \infty$ (no cutoff), but for runs with $u_0 \neq 0$, we used a finite value for K_u (see below). Here, $\alpha_v^{\rm sub}$ is the subinertial range slope and $\alpha_v^{\rm ine}$ is the inertial range slope. For a causal spectrum, as is often used as initial condition for the magnetic field (Durrer et al. 1998; Christensson et al. 2001; Durrer & Caprini 2003), we would use $\alpha_v^{\rm sub} = 3$ in 2D, which results in a δ-correlated magnetic vector potential, but after some time, the slope steepens (see Appendix A). Therefore, we used $\alpha_v^{\rm sub} = 4$, which value is also used in 3D simulations.

The spectrum develops an even steeper part $\propto k^5$ near the peak. Its detailed physical nature has not yet been studied. We note, however, that this effect is reminiscent of the suspected formation of a k^4 bulge in simulations with an initial k^2 spectrum (see Fig. 6 of the supplemental material of Hosking & Schekochihin 2023). So far, however, this has not been observed in such a case (see Reppin & Banerjee 2017 and Brandenburg et al. 2023 for earlier work on 3D nonhelical turbulence with initial subinertial range spectra between k^2 and k^4).

In all cases, we used ${\rm Pr_M} \equiv \nu/\eta = 1$ for the magnetic Prandtl number. The density was initially uniform in all cases and equal to ρ_0. The initial magnetic field strength was such that $v_{\rm A}$ remained subsonic, that is, $v_{\rm A} \ll c_{\rm s}$. This implies that compressibility effects are weak. In most of our runs, we used zero velocity initially, that is, we set $u_0 = 0$ in the expression for $\tilde{\psi}_z(\boldsymbol{k})$ in Sect. 2.2. However, in some cases, we wished to perform a simulation in a larger domain to cover the inverse cascade at later times. In this case, we tested two options. First, we started the simulation using a zero velocity initial condition, which caused an additional adjustment phase, however, during which a nearly self-similarly evolving velocity field built up (run C4). In a second option and to minimize the artifacts resulting from this adjustment phase, we started the run by introducing a nonzero initial velocity with a relative amplitude u_0/B_0 of typically around 0.3 (run C4u) (see Table 1, where we give a detailed list of our simulations).

We solved Eqs. (2)–(4) using the Pencil code (Pencil Code Collaboration 2021), where all relevant diagnostics is computed during run time. Our numerical resolution varied between 1024^2 and 8192^2 mesh points. The lowest wave number in the domain is given by $k_1 = 2\pi/L$, where the domain size L was varied such that the peak wave number k_B fits well into the domain. We used nondimensional quantities by setting $k_B(A) = c_{\rm s} = \rho_0 = \mu_0 = 1$. Here, $k_B(A)$ is the magnetic peak wave number for our reference point A discussed below.

2.3. Diagnostic quantities

We characterized the magnetic field by its mean energy density $\mathcal{E}_{\rm M} = \langle \boldsymbol{B}^2 \rangle/2\mu_0$ and the magnetic integral scale $\xi_{\rm M}$. $\mathcal{E}_{\rm M}(t)$ and $\xi_{\rm M}(t)$ can both be defined in terms of the magnetic energy spectrum $E_{\rm M}(k,t)$, such that $\mathcal{E}_{\rm M} = \int E_{\rm M}\,{\rm d}k$ and $\xi_{\rm M} = \int k^{-1} E_{\rm M}\,{\rm d}k/\mathcal{E}_{\rm M}$. For our reference point A with $k_B(A) = 1$, we then had $\xi_{\rm M} \approx 1$.

Instead of $\mathcal{E}_{\rm M}(t)$, we often used $v_{\rm A}(t)$. Because of mass conservation and since the runs are only weakly compressible, we had $\rho \approx \rho_0$, so $v_{\rm A} \approx \sqrt{2\mathcal{E}_{\rm M}(t)/\rho_0}$ is a good approximation. Moreover, the value of $I_{\rm A} \equiv \langle \rho A_z^2 \rangle/\mu_0\rho_0^2$ is about ${v_{\rm A}}^2{\xi_{\rm M}}^2$; see Table 1. It is well conserved in the limit of large Lundquist numbers, ${\rm Lu} \equiv v_{\rm A}\xi_{\rm M}/\eta$.

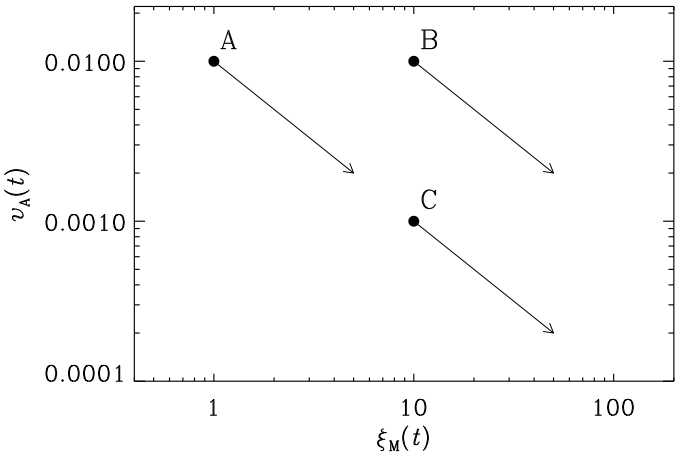


Fig. 1. Sketch of the three initial values of $v_{\rm A}$ and $\xi_{\rm M}$ and their anticipated tracks.

2.4. Simulation strategy

We performed runs such that their initial values of $v_{\rm A}$ and $\xi_{\rm M}$ lay on three points, as sketched in Fig. 1. Points A (run A1) and B (runs B1, B2, and B3) had the same value of $v_{\rm A}$ initially, but different values of $\xi_{\rm M}$, while points B and C (run C1) had the same value of $\xi_{\rm M}$, but different values of $v_{\rm A}$. Furthermore, the initial value of $v_{\rm A}$ for run C1 was such that run A1 was to meet it after some time. We also note that points A and B, which both started at the nominal time $t = 0$, cannot lie on the same isochrone when we expect the isochrone to lie on lines $v_{\rm A} = \xi_{\rm M}/t_*$. Since $I_{\rm A} \sim {v_{\rm A}}^2{\xi_{\rm M}}^2$, points A and C had the same value of $I_{\rm A}$, while the value of point B was 100 times higher because ${\xi_{\rm M}}^2$ is a hundred times larger. The initial points A, B, and C might correspond to different generation mechanisms or even to different model parameters for the same scenario. Examples for the former are generation of magnetic fields during inflation or early Universe phase transitions (e.g., electroweak or QCD).

We recall that in 2D decaying MHD turbulence, $v_{\rm A} \propto t^{-1/2}$ and $\xi_{\rm M} \propto t^{1/2}$ (see also Brandenburg et al. (2024)). Therefore, the evolutionary track is $v_{\rm A} \propto {\xi_{\rm M}}^{-1}$. Thus, if point C has a ten times higher value of $\xi_{\rm M}$ than point A, its value of $v_{\rm A}$ should be ten times lower than for point A. However, to account for a slightly faster decay at small Lundquist numbers, we chose a correspondingly lower initial value of $v_{\rm A}$ in those cases (0.07 for run C1 and 0.09 for runs C4 and C4u). Point B was chosen such that it would lie on the same asymptotic isochrone as point C, that is, $v_{\rm A} = \xi_{\rm M}/t_*$, although the time t_* of this isochrone cannot be the nominal time $t = 0$ chosen in the simulation.

2.5. Alfvén and decay times

The instantaneous Alfvén time is defined as

$$t_{\rm A}(t) = \xi_{\rm M}(t)/v_{\rm A}(t), \quad (6)$$

where $\xi_{\rm M}$ was defined in Sect. 2.3. However, it is in general not true that $t_{\rm A}(t)$ is proportional to t because it depends on an a priori arbitrary shift in the definition of time. (Even the epoch of generation is subject to an arbitrary offset.) We therefore defined

Table 1. Summary of runs with different starting positions given by k_B and B_0.

Run	Lu	k_1	k_B	B_0	u_0	$\eta = \nu$	C_M	t_{offset}	t_{proper}	t_ξ	$t_\mathcal{E}$	$I_A/{v_A}^2{\xi_M}^2$	res.
A1	70	0.01	1	0.1	0	10^{-4}	10.6	420	600	460	440	1.27	1024^2
B1	80	0.001	0.1	0.1	0	10^{-3}	10.6	5700	6000	4600	4400	1.28	1024^2
C1	70	0.001	0.1	0.007	0	10^{-4}	9.4	70 000	63 000	57 000	57 000	1.25	1024^2
B2	190	0.001	0.1	0.1	0	5×10^{-4}	14	6100	5300	5100	5000	1.33	2048^2
B3	540	0.001	0.1	0.1	0	2×10^{-4}	19	8100	6600	6500	6000	1.45	4096^2
A4	1070	0.01	1	0.1	0	10^{-5}	24	1100	740	810	740	1.51	8192^2
C4	1100	0.001	0.1	0.009	0	10^{-5}	22	99 000	74 000	64 000	59 000	1.49	8192^2
C4u	1100	0.001	0.1	0.009	0.003	10^{-5}	22	69 000	76 000	77 000	71 000	1.54	8192^2

Notes. For run C4u, the initial velocity is finite. The anastrophy I_A was defined in Sect. 2.1. The quantities C_M and t_{offset} are defined in Sect. 2.5, t_{proper} is defined in Sect. 3.2, while t_ξ and $t_\mathcal{E}$ are defined in Sect. 3.5.

this shift for each run by computing a linear fit to $t_A(t)$,

$$t_A^{fit}(t) = mt + t_0. \tag{7}$$

Here, t_0 is the run-specific shift, and m is a nondimensional parameter. It would be unity (even at late times) if the decay time is just the Alfvén time, but in general, the actual decay time is longer than $t_A(t)$ by a factor C_M, as discussed in the introduction. It can then be defined as

$$C_M = 1/m, \tag{8}$$

so that

$$t = C_M\, t_A(t). \tag{9}$$

The factor C_M was already determined by Brandenburg et al. (2024) using a different method. As discussed in the introduction, it was found to grow with the Lundquist number and saturate near Lu $\approx 10^4$, independently of the value of Pr_M.

Multiplying Eq. (7) with C_M, we obtained

$$C_M t_A^{fit}(t) = t + C_M t_0 = t + t_{offset}, \tag{10}$$

which shows that $C_M t_0 \equiv t_{offset}$ is the offset time by which we would need to shift the time axis in order to match the proper time of the run.

3. Results

The main results of our simulations are given in Table 1, where we report the values of C_M (which indicates how much longer the decay time is compared to the Alfvén time) as obtained from Eq. (8), along with several other parameters, as explained below. The values of C_M are found to be in the range 10–20. As expected, only for the runs starting at point B do we have a different value of anastrophy (I_A). This is because of the ten times higher value of the eddy magnetic scale, ξ_M. For run B1, the values of the viscosity and magnetic diffusivity were increased by a factor of 10 relative to run A1 because the mesh spacing is here ten times coarser. Since the product ηk_B remained unchanged, the value of Lu is approximately the same. On the other hand, for runs C1 and C4, v_A is about ten times smaller, and even though the mesh spacing is coarser, η and ν were chosen to be the same as for runs A1 and A4, respectively.

Run C4u is similar to run C4, except that the initial value of u_0 was nonzero. We used $u_0/B_0 \approx 0.3$, which reproduces the velocity spectra from the end of run A (see below). For the velocity, we took $k_u = 0.5\,k_B$, $\alpha_{sub}^{(u)} = 2$, $\alpha_{ine}^{(u)} = 0.75$, and $K_v = 8\,k_B$, that is, an exponential cutoff was used to model the end of the velocity inertial range. By contrast, no such cutoff was used for the magnetic field, that is, $K_B \to \infty$.

3.1. Visualizations

We began with visualizations of the z component of the current density J_z at the end of run A4 and the beginning of C4u at the approximate time $t = 10^5$, when the eddy scales of run C4u in the larger domain agreed with those of run A4 in the smaller domain (see Fig. 2). For run C4u, we also show as an inset the result at $t = 5\times10^3$, which is when the turbulent eddies were not yet fully developed. The other two insets show the results for run A4, which is for a smaller domain, and run C4u. In all cases, the insets show a zoomed part of the full domain, and they have the same size.

The images of J_z shown in panels (a)–(c) of Fig. 2 look rather different, although the structures roughly agree in size and proper time. One reason is that our initial conditions have random phases, while for truly turbulent flows, the phases are always correlated with the rest of the field (compare the differences between panels (a) and (b)). Another reason is that J_z develops strong departures from a Gaussian distribution. This is demonstrated in Fig. 3, where we show probability density functions $p(J_z/J_z^{rms})$ for the three insets shown in Fig. 2. For panels (b) and (c), $p(J_z/J_z^{rms})$ develops stretched exponential tails.

3.2. Evolutionary tracks

In Fig. 4 we plot evolutionary tracks for runs A1–C1 and for runs A4 and C4u. We also show isochrones and compare the nominal ones between runs A1 and B1 (dotted lines) with the corrected ones (dashed lines). At early times, the nominal isochrones depart strongly from the asymptotic ones. They are also not parallel to each other. For example, runs A1 and B1 have $t = 0$ as the nominal initial time and would therefore lie on a horizontal line. Qualitatively, this can be explained by the nominal times not describing the proper time, that is, the time that belongs to a given value of ξ_M and v_A, namely

$$t_{proper} = C_M\, \xi_M(0)/v_A(0). \tag{11}$$

In Fig. 4 the blue and black lines, belonging to runs B1 and A1, respectively, have the same value $v_A(0)$ (≈ 0.02), but for run B1, ξ_M is ten times larger than for run A1, and therefore, t_{proper} is also ten times larger (see Eq. (11)). In order for this to be reflected in the time, we expect the shifted time $t + t_{proper}$ for run B1 to be longer than that for run A1. This is indeed the case, and we have $t_{proper} = 6000$ for run B1 and $t_{proper} = 600$ for run A1 (see Table 1).

To draw a revised isochrone for the initial time of these runs, we must connect the initial point of run B1 (with $t_{proper} = 6000$)

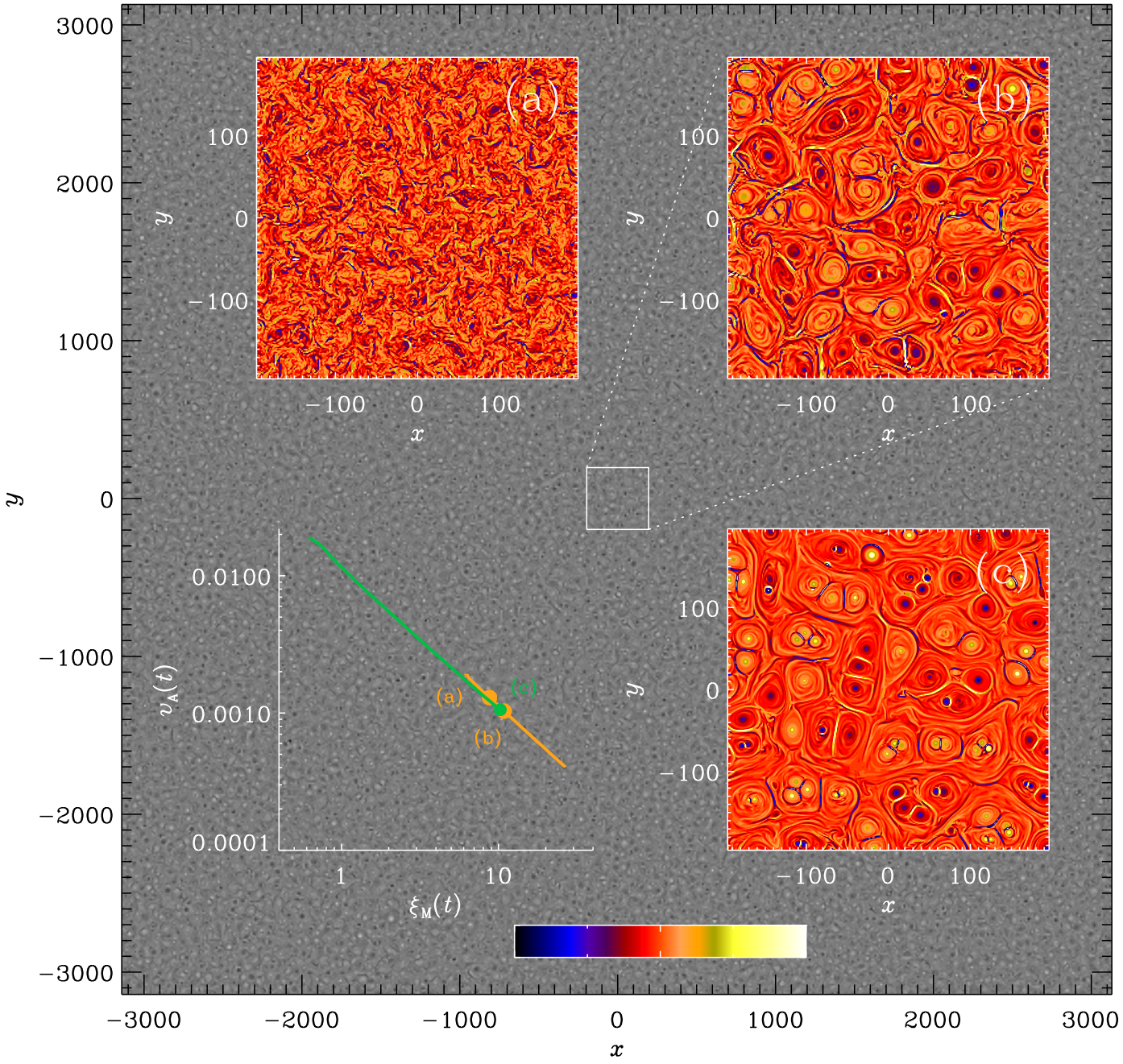


Fig. 2. Grayscale visualization of the current density along the line of sight, J_z, for run C4u at $t = 10^5$. The white square marks the inner part of the domain for which color-scale insets show (a) run C4u at $t = 5 \times 10^3$, (b) run C4u at $t = 10^5$, and (c) run A4 at $t = 2.3 \times 10^5$. The corresponding positions in the ξ_M versus v_A diagram are shown in the lower left inset for (a) and (b) on the track of run C4u in orange and for (c) on the track of run A4 in green. The color table is qualitative and centered symmetrically around zero, corresponding to red shades, with negative and positive extrema corresponding to blue and yellow shades, respectively. The extrema vary between the three panels, so no numbers are given on the color bar.

with the corresponding proper time (5400, which yields 6000 with $t_{\rm proper} = 600$) for run A1. Likewise, for the corrected isochrones for $t_* = 10^3$ and 10^4, we must connect these points of run B1 with those of run A1 at $t = 6400$ (which yields $7000 = 6000 + 10^3$ after adding $t_{\rm proper} = 600$ for run A1) and 15 400 (which yields $16\,000 = 6000 + 10^4$ after adding 600), respectively. This is shown in Fig. 4 (see the dashed lines). The corrected isochrones are now nearly parallel to those at later times. The isochrone for $t_* = 0$ is slightly too steep (see the thicker of the two dashed lines through $t = 0$ for run B1), but this can be explained by statistical errors, predominantly for $t_{\rm proper}$ of run B1. To demonstrate this, we also show as the thinner dashed line the result for the lowest estimate of $t_{\rm proper}$ in Table 1, namely $t_{\mathcal{E}} = 4400$, which yields a line that is now nearly perfectly parallel to the other lines.

We recall that $t_{\rm proper}$ was determined based on the initial values of $C_{\rm M} t_{\rm A}(t)$. As we argued above, it is therefore prone to errors. By contrast, the quantity $t_{\rm offset}$ is based on the vertical offset t_0 of the fit given by Eq. (7) in the graph $t_{\rm A}$ versus t within a given time interval, as we discuss next.

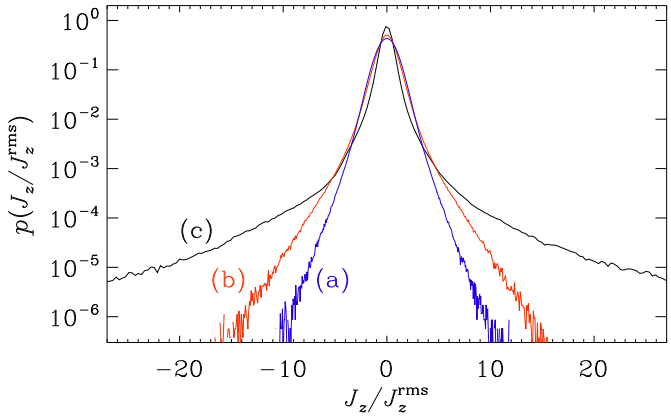


Fig. 3. Probability density functions $p(J_z/J_z^{\rm rms})$ for the three insets shown in Fig. 2.

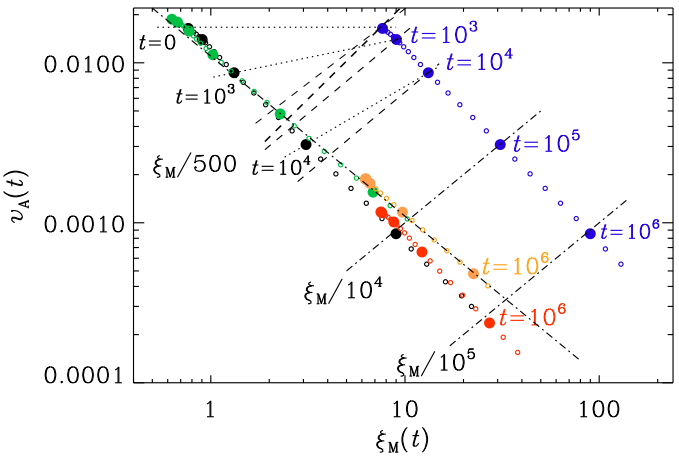


Fig. 4. Evolutionary tracks for runs A1 (black), B1 (blue), and C1 (red), and for runs A4 (green) and C4u (orange). The symbols are logarithmically spaced in time with six open symbols per decade, and each decade is marked with a filled symbol. The dash-dotted lines are Alfvénic isochrones for $t_* = 10^4$ and $t_* = 10^5$. The dotted lines are the nominal isochrones for $t_* = 0$, $t_* = 10^3$, and $t_* = 10^4$ between runs A1 and B1. The dashed lines are the corrected isochrones (see text). To demonstrate the effect of statistical errors, we show two dashed lines through the starting point for run B1 at $t = 0$: for the thicker one, a correction with $t_{\rm proper} = 6000$ was applied, while for the thinner one $t_{\rm proper} = 4400$ was used.

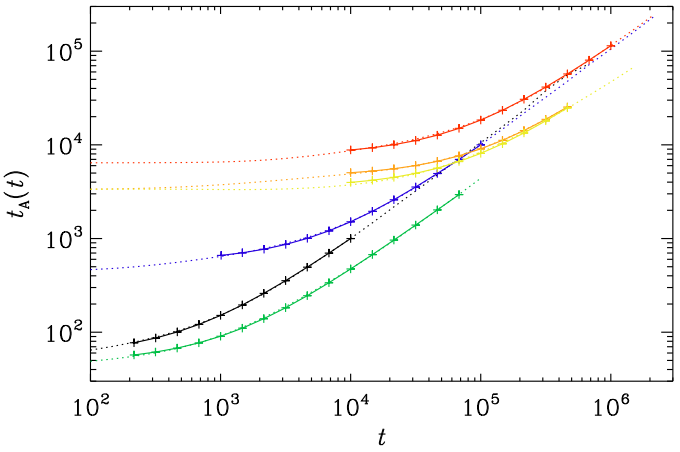


Fig. 5. Linear fits to $t_{\rm A}$ versus t, giving $C_{\rm M} = 1/m$ and t_0 in Eqs. (7) and (8), as noted in Table 1. The color-coding is the same as in Fig. 4. The fits are shown as solid lines, and the original data are shown as dotted lines. In this double-logarithmic representation, the linear fits appear to be curved. The horizontal dotted lines indicate the extrapolation to the asymptote, with the filled symbols marking the values of t_ξ and $t_\mathcal{E}$ given in Table 1.

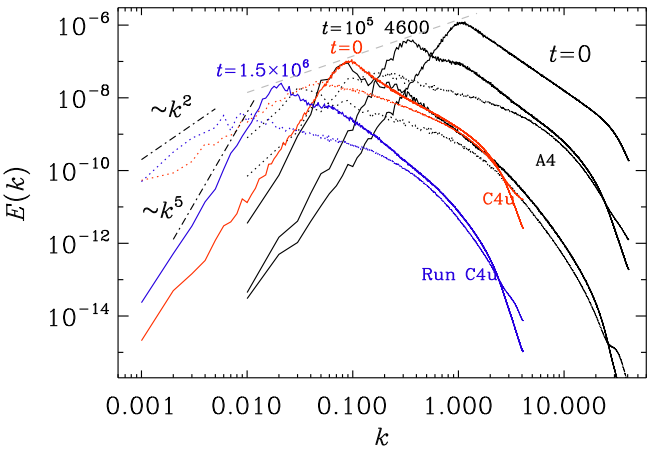


Fig. 6. Magnetic (solid lines) and kinetic (dotted lines) energy spectra for runs A4 and C4u. Run C4u, which is for a ten times larger domain, connects smoothly to the end of run A4. The dashed gray line indicates the expected envelope proportional to k^β with $\beta = 1$.

3.3. Fit to the Alfvén time

In Sect. 3.2 we discussed the usefulness of the quantity $t_{\rm proper}$ as a way of defining a proper initial time. Another possibility is to determine the time t_0 from Eq. (7), as defined in Sect. 2.5. The result is shown in Fig. 5, where we present fits for runs A1, B1, C1, A4, C4, and C4u. A difficulty here lies in the choice of a suitable range over which the fit should be applied. The resulting parameters, which are listed in Table 1, can change somewhat, depending on the start and end time of the fit. Nevertheless, the resulting values of $t_{\rm offset} = C_{\rm M} t_0$ defined in Eq. (10) agree reasonably well with the proper initial time $t_{\rm proper}$ defined in Sect. 3.2, which was based solely on the values of $\xi_{\rm M}$ and $v_{\rm A}$ at the start of the simulation. The differences in the results from the two approaches are therefore expected.

3.4. Spectral evolution

In Fig. 6 we plot spectra from runs A4 and C4u at different times. The general evolution follows the expected behavior for magnetically dominated decaying turbulence (Reppin & Banerjee 2017; Brandenburg et al. 2017, 2024). As time goes on, the spectra shift in an approximately shape-invariant fashion to the lower left. At high wave numbers k, the spectral energy decreases, but it decreases at small wave numbers, which is a clear signature of an inverse cascade. The spectra evolve below an envelope k^β with $\beta = 1$ (see Brandenburg et al. (2024)).

The initial spectrum of run C4u at $t = 0$ approximately also agrees with that of run A4 at $t = 10^5$. This suggests that the start time of run C4u should be shifted by a time span of about $t = 10^5$. Table 1 shows that this agrees with the values we obtained using two different methods.

3.5. Temporal evolution

The theoretically anticipated decay laws for ξ_M and $\mathcal{E}_M$ are

$$\xi_M(t) \approx C_\xi I_A^{1/4} \left(t_\xi + t\right)^{1/2}, \tag{12}$$

and

$$v_A(t) \approx C_{v_A} I_A^{1/4} \, (t_\mathcal{E} + t)^{-1/2}. \tag{13}$$

Qualitatively similar equations were also used by Brandenburg et al. (2024), but we have added the offsets t_ξ and $t_\mathcal{E}$. The coefficients C_ξ and C_{v_A}, from which we obtained the coefficient $C_\mathcal{E} = C_{v_A}^2/2$ for the magnetic energy, are constants that have also been numerically determined in previous works. More specifically, Brandenburg et al. (2024) found $C_\xi \approx 0.13$ and $C_\mathcal{E} \approx 15$ for their cases of 2D decay MHD, and Brandenburg et al. (2025) have compared earlier results for the corresponding 3D cases, where either the mean magnetic helicity density or the Hosking integral is conserved (see Appendix B for the corresponding decay laws). A similarity of the coefficients in different applications might support the idea that these coefficients are universally constant.

Independent estimates for C_ξ and $C_\mathcal{E}$ can be obtained by plotting $\tilde{C}_\xi(t) = \xi_M(t)\, I_A^{-1/4} \left(t_\xi + t\right)^{-1/2}$ and $\tilde{C}_\mathcal{E}(t) = v_A(t)\, I_A^{-1/4} \, (t_\mathcal{E} + t)^{1/2}$, and determining the height of their plateaus at late times (see Fig. 7(a)). The resulting values of C_ξ and $C_\mathcal{E}$ fall into different groups for our small and large Lundquist number runs. For small Lundquist numbers (Lu $\approx$ 70), we have $C_\xi \approx 0.3$ and $C_\mathcal{E} \approx 4$, while for higher values (Lu $\approx$ 1100), we have $C_\xi \approx 0.2$ and $C_\mathcal{E} \approx 10$. These values are higher than those reported in earlier work, but it is possible that the formerly reported values would be approached for larger Lundquist numbers.

At a higher value of Lu, the coefficient C_M in Eqs. (8)–(10) also increases. It can also be obtained, similarly to C_ξ and $C_\mathcal{E}$, as the plateau value of

$$\tilde{C}_M(t;\Delta t) = (t + \Delta t)/t_A(t), \tag{14}$$

provided Δt is close to the correct value. These functions are shown in Fig. 7(b) for different values of Δt. This plot provides not only an independent estimate of C_M for high values of t, but also shows how the evolution of $\tilde{C}_M(t;\Delta t)$ becomes approximately flat for a suitable value of Δt. Comparing with Eq. (7), it is clear that $(t + \Delta t)/t_A(t)$ should become equal to C_M, when $\Delta t = t_0/m = C_M t_0$. This is similar to Eq. (10). Computing $\tilde{C}_M(t;\Delta t)$ therefore yields yet another method for computing C_M and $t_{\rm offset}$ at the same time.

The aforementioned method of extracting a plateau in the evolution of $(t + \Delta t)/t_A(t)$ can provide a more robust technique when the time dependence of $t_A(t)$ is more complicated and simple fits become problematic. This can be the case when determining isochrones from actual magnetogenesis simulations. In that case, we would not simply start from a static snapshot as initial condition, but the magnetic field evolution would include its history toward a particular point. The function $\tilde{C}_M(t;\Delta t)$ might then not yet have reached a plateau. Similarly, at late times, this function can evolve away from the plateau value because of various computational artifacts. So, again, that time stretch would need to be eliminated from the analysis.

To verify the validity of Eqs. (12) and (13), we plot in Fig. 8 the evolution of $\xi_M(t)$ and $\mathcal{E}_M(t)$ for runs A1, B1, and C1. Toward late times, the tracks of runs A1 and C1 agree approximately

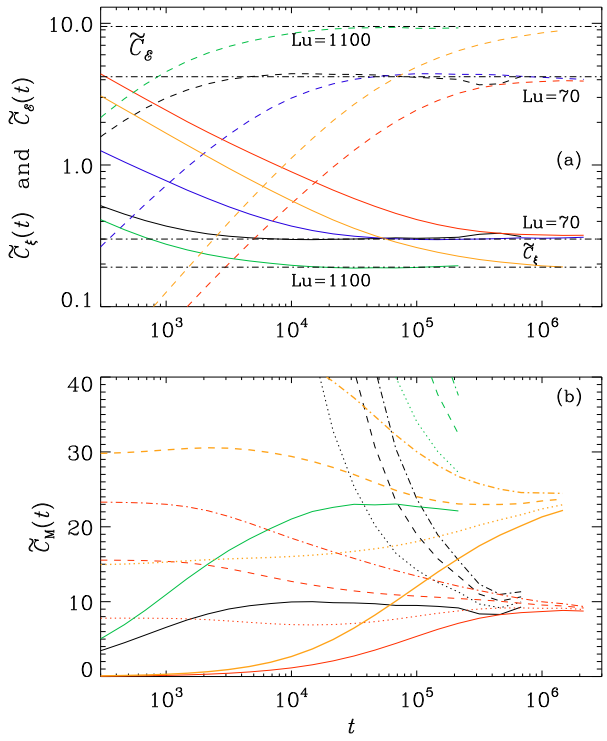


Fig. 7. (a) $\tilde{C}_\xi(t)$ and $\tilde{C}_\mathcal{E}(t)$ for runs A1 (black), B1 (blue), and C1 (red), and for runs A4 (green) and C4 (orange). (b) $\tilde{C}_M(t)$ for runs A1 (black), A4 (green), C1 (red), and C4 (orange). The solid, dotted, dashed, and dash-dotted lines refer to $\Delta t = 0$, 5×10^4, 10^5, and 1.5×10^5, respectively.

because these tracks also agree in Fig. 4, where run C1 continues the evolution of run A1 toward larger scales.

In Fig. 8(a), run B1 is shifted to the left, while in Fig. 8(b), it is shifted to the right. Therefore, run B1 cannot be made to agree with runs A1 or C1 by a shift in the time axis. Instead, Fig. 4 shows that we can collapse the blue line on top of the black and red lines by a 45° shift along the isochrone to the lower left. This corresponds to a combined shift by $1/\sqrt{10}$ in v_A and in ξ_M. Alternatively, we could have plotted $\xi_M/I_A^{1/4}$ and $v_A/I_A^{1/4}$ (see Eqs. (12) and (13)).

For the values given in Table 1, we computed $C_\xi = \min(\xi_M/t^{1/2} I_A^{1/4})$ and $t_\xi = \min(\xi_M{}^2/C_\xi^2 I_A^{1/2})$, while $C_{v_A} = \max(v_A t^{1/2}/I_A^{1/4})$ and $t_{v_A} = \min(C_{v_A}^2 I_A^{1/2}/{v_A}^2)$. The resulting values differ slightly from the intersection points indicated in Fig. 8.

3.6. Scaling exponents

At limited Lundquist numbers and limited scale separation, there is always the possibility of departures from the theoretically expected values. A robust way of inspecting the scaling properties of a run is then to plot the instantaneous scaling exponents

$$q(t) = \mathrm{d}\ln\xi_M/\mathrm{d}\ln t, \quad p(t) = -\mathrm{d}\ln\mathcal{E}_M/\mathrm{d}\ln t. \tag{15}$$

Showing $p(t)$ versus $q(t)$ tends to reveal an evolution along lines of constant $\beta(t) = p(t)/q(t) - 1$ (see Brandenburg & Kahniashvili (2017)). In Fig. 9 we show such a plot for runs A1–C1 and runs A4 and C4u.

While the low Lundquist number runs still deviate quite strongly from the $\beta = 1$ line, the runs with higher values of Lu are much closer to this line. All runs start in the lower left corner at $p = q = 0$ and then settle around the Alfvén line where

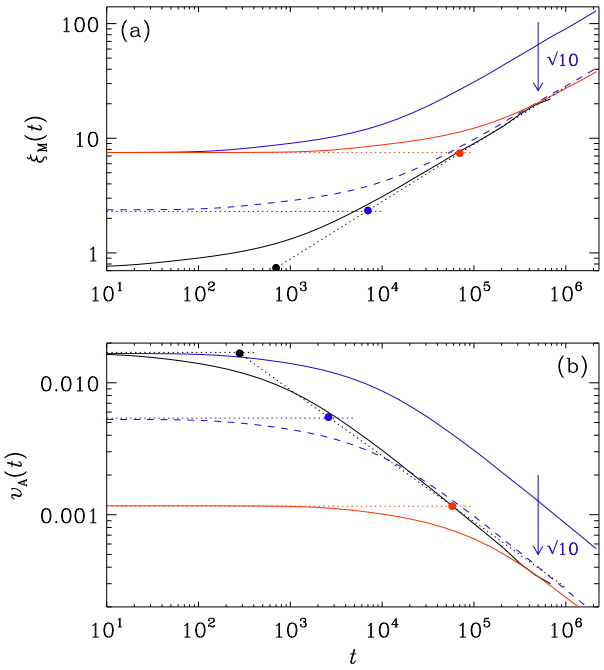


Fig. 8. (a) $\xi_M(t)$ and (b) $v_A(t)$ for runs A1 (black), B1 (blue), and C1 (red). The final stages of run B1 (blue) can be collapsed onto those of runs A1 (black) and C1 (red), as indicated by the blue dashed line. The horizontal dotted lines indicate the extrapolation to the asymptote with the filled symbols marking the values of t_ξ and $t_\mathcal{E}$.

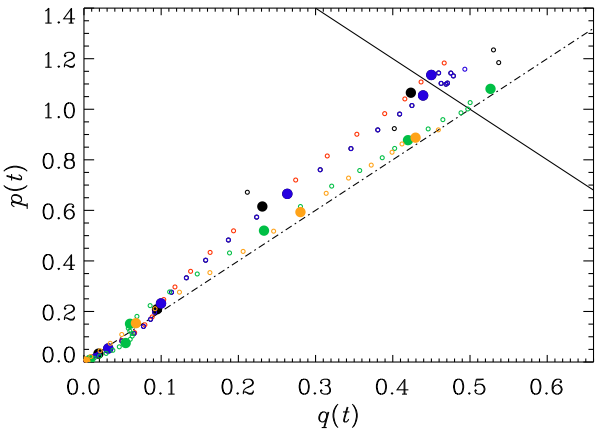


Fig. 9. Evolution in the pq diagram for the same runs as in Fig. 4, i.e., for runs A1 (black), B1 (blue), and C1 (red), and for runs A4 (green) and C4u (orange). Again, the symbols are logarithmically spaced in time with six open symbols per decade, and each decade is marked with a filled symbol. The dash-dotted line has the slope $\beta + 1 = 2$, and the solid line corresponds to the Alfvén line where $p = 2(1 - q)$.

$q = 1/2$ and $p = 1$. This is analogous to earlier findings for 3D turbulence (Brandenburg & Kahniashvili 2017).

3.7. Conservation of anastrophy

Because the Lundquist number is finite, the anastrophy will not be perfectly conserved. To quantify the degree of anastrophy conservation, we determined the decay exponent in I_A with time. For a given Lundquist number, we found that $I_A \propto t^{-s}$. We determined the value of s for runs A1, B2, B3, and A4, which cover

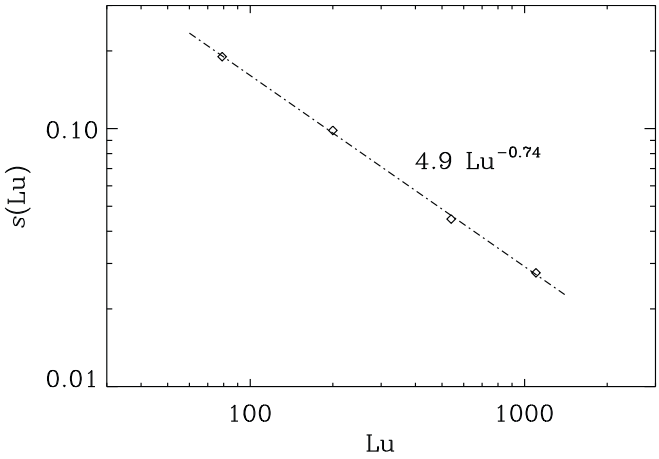


Fig. 10. Degree of anastrophy conservation quantified by the decay exponent s as a function of Lu.

the range $80 \le \mathrm{Lu} \le 1100$. We found that $s \ll 1$, which suggests that the decay is slow.

It is important, however, that the value of the exponent s diminishes with increasing value of the Lundquist number. The results are plotted in Fig. 10, and $s \approx 3.1\,\mathrm{Lu}^{-2/3}$. Thus, for $\mathrm{Lu} = 10^3$, we find $s = 0.03$. This would mean that the anastrophy would decay by a factor of 3 after 16 orders of magnitude in time, which is the range expected in cosmological applications. For $\mathrm{Lu} = 10^6$, however, the anastrophy would decay by just 2% after 30 orders of magnitude in time.

A similar analysis was previously conducted for the Hosking integral (Zhou et al. 2022), and the decay exponent was found to decrease with Lundquist number approximately like $0.8\,\mathrm{Lu}^{-1/4}$. To the best of our knowledge, similar decay laws have not been discussed before. It is therefore still unclear how general our results are.

3.8. Dependence on the initial velocity field

We examined the dependence of the evolutionary tracks on the presence of an initial velocity field. In Fig. 11 we compare models with different initial values of u_0/B_0. For $u_0 = 0$, the tracks show a small upward trend in the ξ_M versus v_A diagram (see the black line in Fig. 11). When $u_0/B_0 \approx 0.3$, the track is straighter. Finally, when $u_0/B_0 > 0.3$, the track evolves first to the left, that is, the magnetic length scales initially decrease, and the track later develops a downward trend before entering the expected decay behavior. This is qualitatively explained by the effect of turbulent diffusion exerted by the excessive strength of the flow compared to the flow amplitude from the magnetic driving alone.

It is possible that for very large Lundquist numbers, a strong velocity field not only enhances the decay of the magnetic field, but leads to additional dynamo action and thereby to an enhancement of the magnetic field strength. Dynamo action is only possible in 3D, but even in 2D, there can be some transient enhancement. However, to show this more clearly, we would need to start with a magnetic field whose Alfvén velocity is lower than that of the velocity field. This is shown Fig. 12, where we have lowered the magnetic field strength by a factor 100.

That the magnetic field enhancement is stronger for weaker than for stronger fields was quite clearly shown in the 3D decay simulations of Brandenburg & Ntormousi (2025). The magnetic field grew during the early decay phase, when the magnetic Reynolds number based on the decaying velocity field was still

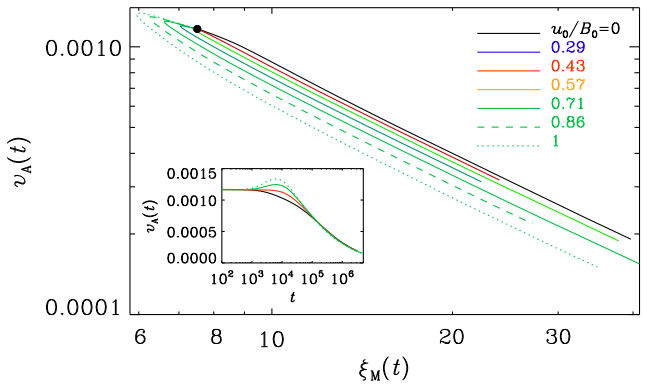


Fig. 11. $v_A(t)$ versus $\xi_M(t)$ for different values of u_0/B_0. The starting point is indicated by a filled black symbol. The black line is for $u_0 = 0$ and shows a slight upward bend, while the blue line for $u_0/B_0 = 0.29$ is nearly straight. For higher values of u_0/B_0, the track first moves to the upper left corner, corresponding to a slight increase in field strength (shown in the inset) and a smaller typical length scale, before entering the usual decay track at slightly smaller field strength. The insect only shows the lines for $u_0/B_0 = 0$, 0.43, 0.71, and 1.

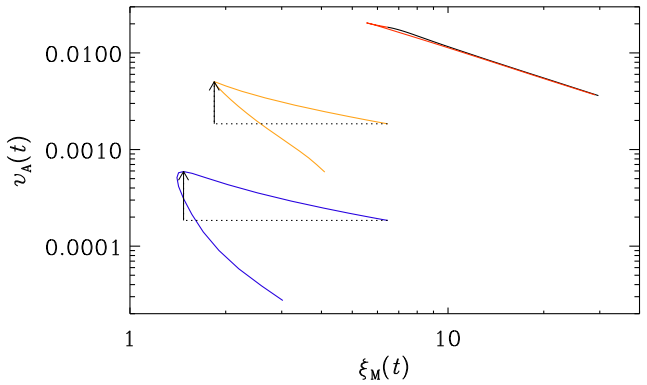


Fig. 12. $v_A(t)$ versus $\xi_M(t)$ for run B3 with $u_0 = 0$ (black) and 0.05 (red), both with $B_0 = 0.1$, and $B_0 = 10^{-2}$ (orange) and $B_0 = 10^{-3}$ again with $u_0 = 0.05$ (blue). The vertical arrows indicate the increase in v_A from the initial value to the maximum.

large enough. The growth was less pronounced when the initial magnetic field was stronger, which simply led to an earlier saturation. The simulations shown in Fig. 12 display a similar trend, even though the growth in this case was not caused by a dynamo.

3.9. Development of the inertial range

In all the cases discussed above, we assumed that the initial magnetic field is described by a power-law spectrum with an inertial range with a spectral slope $\alpha_B^{\rm ine} = -2$. However, the initial magnetic field might have a very different spectrum, corresponding to a different magnetogenesis scenario. In order to assess the effect of a mismatch between the initial spectrum and the spectrum that develops in the later self-similar phase, we considered a case with a very short inertial range. For $\alpha_B^{\rm ine} \to -\infty$, the spectrum would decay almost instantaneously with k. In practice, we chose $\alpha_B^{\rm ine} = -20$, so that the inertial range was almost absent. The comparison with the original run C4 is shown in Fig. 13. With a significantly shorter inertial range, the length scale ξ_M is somewhat larger, and the initial field strength is slightly weaker. Nevertheless, the initial values of ξ_M and v_A still lie on a similar track as for the original run C4. After a short time, however, an inertial range develops at the expense of the magnetic energy near the spectral peak. This leads to a drop in the field strength. After this, however, the point (ξ_M, v_A) continues on a new track that corresponds to a track with a 1.7 times smaller anastrophy. This is because the initial condition already had a lower initial value of the anastrophy. This demonstrates the importance of characterizing the initial condition not just by their initial values of ξ_M and v_A, but also by the value of the relevant conserved quantity.

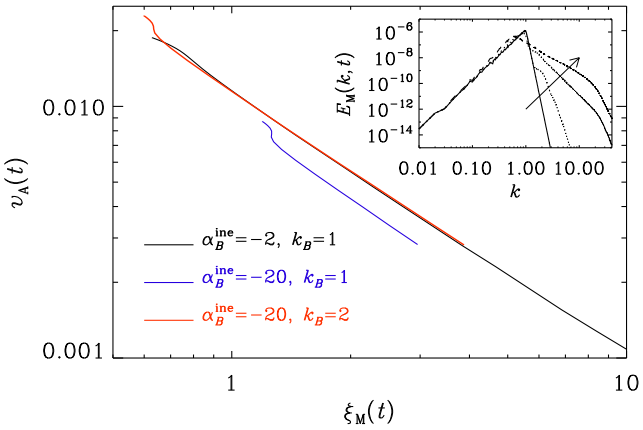


Fig. 13. $v_A(t)$ versus $\xi_M(t)$ for run C4 with $\alpha_B^{\rm ine} = -2$ (the usual case, black) and $\alpha_B^{\rm ine} = -20$ (red). The inset shows the build-up of the inertial range at times $t = 0$ (solid line), $t = 100$ and 300 (dotted lines), and $t = 1000$ (dashed line), corresponding to 0, 3, 8, and 11 Alfvén times.

To compensate for the larger initial length scale for runs with $\alpha_B^{\rm ine} = -20$, we also compare our results in Fig. 13 with a run with $k_B = 2$ instead of $k_B = 1$. Furthermore, to adjust for the resulting drop in the anastrophy, we increased the initial field strength by a factor of 3.7. In this case, the trajectory matches that for $\alpha_B^{\rm ine} = -2$ and $k_B = 1$ reasonably well. In Fig. 13 we also show the build-up of the inertial range at subsequent times $t = 100$, 300, and 1000, corresponding to $t/t_A \approx 3$, 8, and 11 Alfvén times. (The values of $t_A = \xi_M/v_A$ itself have slightly increased with time because ξ_M has increased and v_A has decreased, giving $t_A = 31$, 41, and 87 for these three times.) Thus, it takes at least ten Alfvén times to establish an inertial range. This is consistent with recent requirements found by Silver et al., (in prep.) for magnetic field amplification in young galaxies.

4. Conclusions

One possibility to explain the origin of extragalactic magnetic fields is that volume-filling seed magnetic fields were generated by primordial processes in the early Universe, for instance, during or after inflation, or during the electroweak or QCD phase transitions (see Subramanian (2016) and Vachaspati (2021) for recent reviews). In each of these cases, it is crucial to understand how the magnetic field characteristics could have been modified by cosmic evolution in order to match their evolved configurations with possible constraints from observations of the local Universe (e.g., Neronov et al. 2024; Carretti & Vazza 2025). The goal of this work was to derive universal isochrones that rule the evolution of primordial magnetic fields since their generation.

As the primordial magnetic field decays due to the turbulence driven by the magnetic field itself, its characteristic length scale increases in a predictable way until it reaches the line $v_A \sim \xi_M/t_*$ in the diagram of Alfvén speed versus length scale

(Banerjee & Jedamzik 2004). Here, t_* is the time today, that is, about 14 Gyr. This line is the final isochrone. At earlier times, that is, for lower values of t_*, we expect there to be similar isochrones that are all parallel to each other. However, for very low values of t_*, it becomes important to know how exactly the time is defined. We have proposed different methods that give broadly similar results (see Table 1).

It might have been thought that not all conceivable choices of initial field configurations are permissible and that their length scales must be compatible with what can be explained in terms of the largest processed eddy that can develop in the time available. This is not the case, however. Instead, any combination of initial values of ξ_M and v_A, or k_B and B_0, is allowed. All that is needed is to reset the time axis such that t agrees with $C_M t_A$. This was demonstrated in Sect. 3.2, where we defined t_{proper} as a proper initial time of each simulation. The other measures of t_{proper} can only be determined a posteriori after C_M and t_A have been determined. This is not a restriction on the overall idea, however.

There is always the possibility of artifacts in simulating decaying turbulence that we should keep in mind. First of all, we showed in Sect. 3.2 that the expected evolutionary tracks are only obtained for high enough values of Lu. This also means that the resolution, that is, the number of mesh points N, should be high enough. However, there is an important compromise to strike between a small peak wave number k_B, which would increase the value of Lu, and a low value of k_1, which is needed to capture a sufficiently long subinertial scale range. These factors can affect the determination of isochronous in unexpected ways.

Several possible tools could have been employed to optimize the use of the available finite resolution. One tool is the use of time-dependent viscosities and magnetic diffusivities, and another tool is the use of hyperviscosities and magnetic hyperdiffusivities (Hosking & Schekochihin 2021). Their effects have, to some extent, already been examined in Zhou et al. (2022). We did not use these tools. However, it would be interesting to determine isochrones with these different tools to optimize the use of the available resolution.

In agreement with earlier findings, a given magnetic energy spectrum always implies a certain kinetic energy spectrum. Starting with zero initial velocity always leads to an initial adjustment phase during which the velocity field reaches its asymptotic self-similar state. It is characterized by a peak at a wave number that is about half the wave number of the magnetic peak and a hight that is about one-third of the magnetic peak, followed by a spectrum with a slope $\sim k^{-3/4}$. The reason for this empirically determined spectrum is still unclear.

The main purpose of our work was to define proper offset times, such as t_{proper}, t_{offset}, t_ξ, $t_\mathcal{E}$, and Δt, defined in Sects. 3.3 and 3.5. There, we discussed the differences in the results from the five definitions. The differences were relatively small, and it is difficult to say which of the five methods is the best. The ultimate test would be the application to early Universe magnetogenesis mechanisms, when the magnetic field has just entered the radiation-dominated era. In this case, the temporal evolution can be very complex so that a robust method must be applied. This might well be the determination of Δt from a plateau in the evolution of $\tilde{C}_M(t;\Delta t)$.

Data availability

The source code used for the simulations of this study, the PENCIL CODE (Pencil Code Collaboration 2021), is freely available on https://github.com/pencil-code/ with its latest developments. The DOI of the code is https://doi.org/10.5281/zenodo.2315093. The simulation setups and corresponding data are freely available on DOI:10.5281/zenodo.20603413, as well as on norlx65.nordita.org/ brandenb/projects/Isochrones.

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Appendix A: Effect of initial subinertial range slope

As we have noted in Sect. 2.2, the subinertial range spectrum is not consistent with a causal one, which would be $\propto k^3$ in 2D. Instead, there is a trend to develop a steeper one. In most of our runs, we have therefore used an initial k^4 spectrum for the magnetic field. In this appendix, we show that the effect of the initial slope on the evolutionary track is negligible; see Fig. A.1 where we compare the tracks for $\alpha_B^{\rm sub} = 3$, 4, and 5.

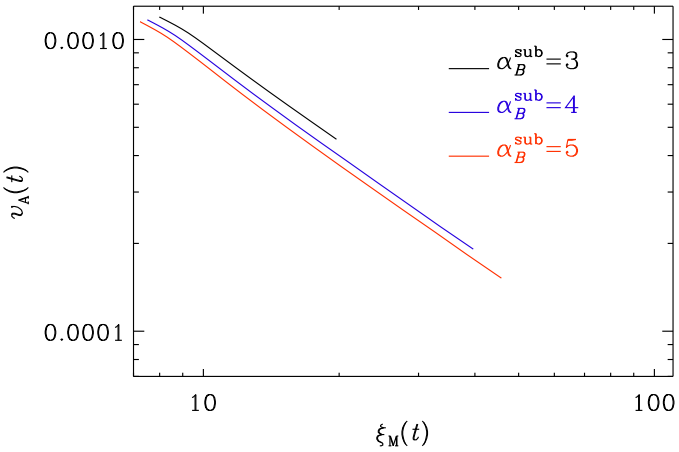


Fig. A.1. Effect of different slopes $\alpha_B^{\rm sub} = 3$, 4, and 5.

Table A.1. Connection between initial and final subinertial range slopes, compiled from BK17 (Brandenburg & Kahniashvili 2017), BSV23 (Brandenburg et al. 2023), and BK26 (Brandenburg & Kumar 2026).

Case	initial	final	reference
3D helical MHD	k^2	k^4	BK17
3D helical MHD	k^6	k^4	BK17
3D nonhelical MHD	k^2	k^2	BSV23
2D Hall cascade	k^4	k^5	BK26

In Table A.1, we summarize some related results from earlier work. In particular, for helical magnetic fields, Brandenburg & Kahniashvili (2017) found that an initial k^2 spectrum steepens to a k^4. Conversely, a steeper initial k^6 becomes shallower, and is then again consistent with a k^4 spectrum. On the other hand, for 3D nonhelical MHD, an initial k^2 spectrum remains $\propto k^2$ (Brandenburg et al. 2023).

During the time span when the magnetic field is still being generated, and it grows exponentially in time around a particular wave number k_*, the spectrum for $k < k_*$ can have a slope steeper than $\propto k^4$. An example is given by the chiral magnetic effect, where the magnetic energy spectrum was found to be $\propto k^{16}$; see Fig. 2 of Brandenburg et al. (2017). This was argued to be caused by mode coupling, which affects the magnetic energy spectrum even at very small values of k.

The fact that the spectrum can instantaneously develop magnetic energy at very small k is a consequence of the Fourier transform: the Fourier decomposition of any spatially compact field configuration can contain modes at low k values. This should not be confused with a superalfvénic or even superluminal evolution of magnetic fields at large scales.

Appendix B: Decay laws in 3D

In Sect. 3.5, we presented the decay laws for $\xi_{\rm M}$ and $\mathcal{E}_{\rm M}$ for the 2D case when the anastrophy is conserved. This is not applicable to 3D, when the local magnetic helicity density is nonvanishing (even if it vanishes on average). In that case, the Hosking integral $I_{\rm H}$ is conserved. It characterizes the magnetic helicity fluctuations (Hosking & Schekochihin 2021; Zhou et al. 2022) and has dimensions of cm^9 s^{-4}. Therefore, instead of Eqs. (12) and (13), we now have

$$\xi_{\rm M}(t) \approx C_\xi^{\rm H} I_{\rm H}^{1/9} \left(t_\xi^{\rm H} + t\right)^{4/9}, \tag{B.1}$$

and

$$v_{\rm A}(t) \approx C_{v_{\rm A}}^{\rm H} I_{\rm H}^{1/9} \left(t_{\mathcal{E}}^{\rm H} + t\right)^{-5/9}, \tag{B.2}$$

where $t_\xi^{\rm H}$ and $t_{\mathcal{E}}^{\rm H}$ are the corresponding offsets. This should be compared with the case when the average of the magnetic helicity density, $I_{\rm M}$, is nonvanishing. It has dimensions of cm^3 s^{-2} and in that case, we have

$$\xi_{\rm M}(t) \approx C_\xi^{\rm M} I_{\rm M}^{1/3} \left(t_\xi^{\rm M} + t\right)^{2/3}, \tag{B.3}$$

and

$$v_{\rm A}(t) \approx C_{v_{\rm A}}^{\rm M} I_{\rm M}^{1/3} \left(t_{\mathcal{E}}^{\rm M} + t\right)^{-1/3}, \tag{B.4}$$

where $t_\xi^{\rm M}$ and $t_{\mathcal{E}}^{\rm M}$ are the corresponding offsets. The transition from one to the other decay law depends on the relative strengths of $I_{\rm H}$ and $I_{\rm M}$, and the values of the nondimensional coefficients $C_\xi^{\rm H} \approx 0.14$, $C_\xi^{\rm M} \approx 0.12$, and $C_{v_{\rm A}}^{\rm H} \approx C_{v_{\rm A}}^{\rm M} \approx 4.0$; see Eqs. (3.8) and (3.9) of Brandenburg & Banerjee (2025).